

OSI Model in Networking: OSI Model's Impact and Involvement in Networking

Liela Pressley

College of Science, Engineering, and Technology, Grand Canyon University

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Instructor Joel Short

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The OSI model is a reference/logical model used to describe the functionality of the communications between systems. For networking, information security, and other technology professionals, this model is used to understand and organize the various moving parts and devices involved in the Network. From the hardware to the User Interface, communication, data packets, and security are interconnected and organized by Layers. This paper will discuss how OSI model layers affect the network, how it is useful in troubleshooting, and two different types of Local Area Networking technologies.

How OSI Model layers Affect Networking and Its Design

The OSI model is used to separate network communication into simplified layers. In implementing individual OSI layer functionalities, various vendors and their equipment interoperate consistently and efficiently. By adhering to the OSI model, network designs have the ability to be updated and innovated with new functionalities because it is easier to add new protocols and services to a layered architecture (*Mitchell, 2019*). The organization and guided rules of protocols transform data streams between levels and are run by detailed interoperating designs (*Orzen, 2014*). Understanding the interaction within the OSI model functionality of networks is achieved with case studies. By utilizing the layered structure different manufacturers are able to design specific applications that communicate appropriately and hardware without disturbing the entire network stack. The OSI model reduces complexity, provides a standard for all interfaces, simplifies teaching/understanding, and allows for modular engineering (*ComputerNetworkingNotes, 2021*). All of these different characteristics allow for a smooth, consistent, and efficient design process.

LAN Switching Technologies and the OSI Model

A Local Area Network (LAN) is responsible for data communication across various terminals and computers in a specific area or building. There are two major types of LAN switching technology, Wired LAN and Wireless LAN. Within those two categories, there are various LAN technologies, an example of Wireless LAN is Wi-Fi while Wired technologies include, Ethernet and Switch.

WiFi, also known as Wireless Fidelity, uses radio waves to provide wireless Internet and network connections which are established using wireless adapters to create hotspots. Devices connect to the Internet via a WLAN network and a wireless access point. Each WLAN has an access point that is responsible for receiving and transmitting data. Ethernet, which is widely used in LAN technology is a low-cost network implementation that requires lots of cabling but is known for its easy maintenance, flexibility, low costs, and understandable features. Ethernet operates within the Physical and Data Link Layers of the OSI Model (*GeeksforGeeks, 2021a*). A network switch, segments the networks into different subnetworks called LAN segments while filtering/forwarding packets between LAN segments based on MAC addresses (*GeeksforGeeks, 2021*).

When referencing the OSI Model, the Local Area Network is found within the Data Link layer (2). This layer is responsible for data transfer between network nodes as well as the delivery of frames between devices within the Local Area Network. Unlike Wide Area networks, data link frames within the LAN do not cross boundaries, and focus on local delivery, addressing, and more (*GeeksforGeeks, 2022*).

How the OSI Model Benefits Troubleshooting

Though TCP/IP is widely adopted, OSI is a great reference for troubleshooting within the Network. The more detailed model is referred to by Networking and Security professionals for various purposes within troubleshooting. Some utilize the OSI model to distinguish jurisdiction/responsibility amongst employees, while others use it to specify where a problem is located (*NetworkChuck, 2020*). Using an image of the OSI model is helpful in visualizing the various parts of the data paths which allows for easier problem analysis and solving. There are different methods of troubleshooting such as starting from Layer 1 up to Layer 7, starting from Layer 7 and working down to Layer 1, or starting at a specific layer and then moving in the direction that is best for the issue at hand (*Davis, 2009*). Having a written, knowledgeable or visual representation of the OSI model is very important to many troubleshooting occurrences within Networking and Security.

The OSI model is essential to the development, innovation, and usage of the Network and its designs. Networking and tech professionals reference the model in order to properly maintain the data transmissions, protocols, design, and troubleshoot in a cohesive and efficient manner. The Data Link Layer of the OSI model is generally responsible for data transfer and frames within the Local Area Network. Wired and Wireless network technologies are used and have different functionalities and benefits. In sum, the OSI model is a necessary tool and framework within Networking and its various components.

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